



The Energy University

CSEB3313 OBJECT ORIENTED PROGRAMMING

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Documentation for the Library Management System

Purpose:

This system is designed to manage items in a library, allowing for the addition, removal, searching, borrowing, and returning of items. It also includes functionality for tracking the availability of items and calculating fines for late returns.

Classes:

1. LibraryItem (Base Class)

- **Attributes:**
 - itemID (int)
 - title (String)
 - author (String)
 - isAvailable (boolean)
- **Methods:**
 - **Constructor:** Initializes itemID, title, author, and sets isAvailable to true.
 - **Getters:**
 - getItemID(): Returns the itemID.
 - getTitle(): Returns the title.
 - getAuthor(): Returns the author.
 - isAvailable(): Returns the availability status.
 - **Setters:**
 - setAvailable(boolean available): Sets the availability status.
 - **Abstract Methods:**
 - getType(): Returns the type of the item ("Book", "Magazine", "Thesis").
 - displayInfo(): Displays the details of the item.

2. Book (Subclass of LibraryItem)

- **Attributes:**
 - genre (String)
 - ISBN (String)
- **Methods:**
 - **Constructor:** Initializes itemID, title, author, genre, and ISBN.
 - **Override Methods:**
 - getType(): Returns "Book".
 - displayInfo(): Displays detailed information of the book, including availability status.

3. Magazine (Subclass of LibraryItem)

- **Attributes:**
 - issueNumber (int)
 - publishedDate (String)
- **Methods:**
 - **Constructor:** Initializes itemID, title, author, issueNumber, and publishedDate.
 - **Override Methods:**
 - getType(): Returns "Magazine".
 - displayInfo(): Displays detailed information of the magazine, including availability status.

4. Thesis (Subclass of LibraryItem)

- **Attributes:**
 - researcherName (String)
 - supervisor (String)
- **Methods:**
 - **Constructor:** Initializes itemID, title, author, researcherName, and supervisor.
 - **Override Methods:**
 - getType(): Returns "Thesis".
 - displayInfo(): Displays detailed information of the thesis, including availability status.

5. Library

- **Attributes:**
 - libraryItems[] (LibraryItem[])
 - itemCount (int)
- **Methods:**
 - **Main Method:** The system prompts users to choose from various operations like displaying items, searching items, borrowing, and returning items.
 - **displayAllItems():** Displays all items currently available in the library. If no items are available, it will notify the user.
 - **searchItems(Scanner scanner):** Allows the user to search items by either title or type. Displays matching items based on the query.
 - **borrowItem(Scanner scanner):** Allows the user to borrow an item by providing the itemID. If the item is available, it updates the item's availability status and calculates due date and fines.
 - **returnItem(Scanner scanner):** Allows the user to return an item by providing the itemID. If the item is returned late, it calculates fines based on a predefined late period.

- **addNewItem(Scanner scanner):** Allows the user to add a new item (Book, Magazine, or Thesis) to the library. It prompts for the necessary details and adds the item to the libraryItems array.
- **removeItem(Scanner scanner):** Allows the user to remove an item from the library by specifying the itemID. It shifts the remaining items in the array to maintain the list's order.
- **addItem(LibraryItem item):** Adds a new LibraryItem to the libraryItems array if space is available. If the library is full, with a maximum capacity of 10 items, it displays an error message.
- **calculateFines(int lateDays):** Calculates the fine for late return based on the number of late days. The fine is calculated as RM0.50 per day.
- **calculateDueDateAndFines():** Placeholder method to calculate the due date and fines for borrowed items.

Sample Test Cases:

Test Case 1:

```
Library System
1. Display all items
2. Search items by title or type
3. Borrow an item
4. Return an item
5. Add an item
6. Remove an item
7. Exit
Enter your choice: 2

Search for items in the library
You can search by either 'title' or 'type'.
Enter your choice ('title' or 'type'): type
Enter the type of the item (Book, Magazine, Thesis): Book
Book: The Great Gatsby by F. Scott Fitzgerald, Genre: Fiction, ISBN: 9780743273565
```

The user selects the second option from the library system to search for items by title or type. Upon entering the item type, 'book,' the system displays the full details of available books.

Test Case 2:

```
Library System
1. Display all items
2. Search items by title or type
3. Borrow an item
4. Return an item
5. Add an item
6. Remove an item
7. Exit
Enter your choice: 3

Enter item ID to borrow: 2
Item borrowed successfully.
Please return the item within 7 days.
```

The user selects the third option from the library system, which is to borrow an item. The system prompts the user to enter the item ID. After user enters a valid id, a confirmation message is displayed, indicating that the borrowing process was successful and that the item must be returned within the allocated time period.

Test Case 3:

```
Library System
1. Display all items
2. Search items by title or type
3. Borrow an item
4. Return an item
5. Add an item
6. Remove an item
7. Exit
Enter your choice: 4

Enter item ID to return: 1
Item returned successfully.
Late return! The fine is: RM3.5
```

The user selects the fourth option from the library system, which is to return an item. The system prompts the user to enter the item ID. After user enters a valid ID, a confirmation message is displayed, indicating that the return process was successful but that the item was returned after the due date. Consequently, a fine is imposed.

Test Case 4:

```
Library System
1. Display all items
2. Search items by title or type
3. Borrow an item
4. Return an item
5. Add an item
6. Remove an item
7. Exit
Enter your choice: 5

Enter item type (book/magazine/thesis): thesis
Enter item ID: 3
Enter title: How to be a comedian
Enter author: Mr Bean
Enter researcher name: Teddy
Enter supervisor: The Pink Aunty

Library System
1. Display all items
2. Search items by title or type
3. Borrow an item
4. Return an item
5. Add an item
6. Remove an item
7. Exit
Enter your choice: 2

Search for items in the library
You can search by either 'title' or 'type'.
Enter your choice ('title' or 'type'): title
Enter the title of the item: How to be a comedian
Thesis: How to be a comedian by Mr Bean, Researcher: Teddy, Supervisor: The Pink Au
```

The user selects the fifth option from the library system, which is to add an item. The system prompts the user to enter the item type, item ID, title, and author for all items. For the thesis, it also requires the researcher's name and supervisor. The item is then automatically added to the system. To verify that the item has been added, the user selects the second option, which is to search for items. The system displays the details of the newly added item.

Test Case 5:

```
Library System
1. Display all items
2. Search items by title or type
3. Borrow an item
4. Return an item
5. Add an item
6. Remove an item
7. Exit
Enter your choice: 6

Enter item ID to remove: 3
Item removed successfully.
```

The user selects the sixth option from the library system, which is to remove an item. The system prompts the user to enter the item ID of the item to be removed. In this case, the user wants to remove an existing item in the system, which is a thesis with item ID 3. After the item is removed, the system displays a message confirming that the item has been successfully removed.